MIL-G-52627A(ME)
27 April 1971
SUPERSEDING
MIL-P-52627(ME)
23 September 1968

MILITARY SPECIFICATION

GENERATOR SET, GASOLINE ENGINE, TRAILER MOUNTED,

5 KW, 60 HERTZ, PU-409A/M

1. SCOPE

1.1 This specification covers a trailer mounted generator set (hereinafter referred to as a power unit) designated PU-409A/M. The power unit consists of a military design, gasoline-engine-driven, 5 kw, 60 Hz, generator set, various auxiliary equipment, brackets, and hardware mounted on a cargo trailer.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin.
FF-N-105	- Nails, Wire, Brads, and Staples.
NN-P-530	- Plywood, Flat Panel.
QQ-S-781	- Strapping, Steel, Flat and Seals.
UU-T-81	- Tags, Shipping and Stock.
PPP-B-591	- Boxes, Fiberboard, Wood-Cleated.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Box, Fiberboard.
PPP-F-320	- Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes.
PPP-P-40	- Packaging and Packing of Hand Tools.
PPP-T-60	- Tape: Pressure-Sensitive Adhesive, Waterproof, for Packaging.
PPP-T-97	- Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

FSC 6115

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Military

MIL-P-116	- Preservation, Methods of.
MIL-B-131	- Barrier Material, Water Vaporproof, Flexible, Heat-Sealable.
MIL-G-12803	- Gasket Material, Non-Metallic.
MIL-L-21260	- Lubricating Oil, Internal Combustion Engine, Preservative and Break-In.
MIL-G-52684	- Generator Sets, Gasoline Engine Driven, 0.5 KW Thru 10 KW, 60 Hertz, 400 Hertz and 28 Volt DC, Packaging of.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-281	- Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies: Preservation and Packaging of.
MS16992	- Bolt, (Screw), Lag, Hex Head, Gimlet Point.
MS27183	- Washer, Flat-Round, Steel, Cadmium Plated, General Purpose (In/MM).

DRAWINGS

ME

DL13214E1224	- Power Unit, PU-409A/M.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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U. S. DEPARTMENT OF COMMERCE

Commercial Standard CS 35	- Hardwood Plywood.
Products Standard PS 1	- Softwood, Plywood; Construction and Industrial.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

A283	- Low and Intermediate Tensile Strength Carbon Steel Plates of Structural Quality.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 Description. The power unit shall be in accordance with Data List DL13214E1224 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are engineering design drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The supplier shall furnish a power unit for examination within the time frame specified, to prove prior to starting production, that his production methods will produce power units that comply with the requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified on the applicable drawings.

3.4 Government-furnished property. The following property in the quantities indicated will be furnished by the Government (see 6.4):

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Item	Description	Identification	Qty for each PU-409A/M
1	Trailer, cargo, 3/4 Ton, 2 wheel, M101A1	MIL-T-10579	1
2	Generator Set, Gasoline-Engine-Driven, 5 KW, 60 hertz.	FSN 6115-017-8240	1

3.5 Workmanship. Workmanship shall be of a quality to assure that the power unit is free from defects resulting from defective material, incorrect manufacturing or assembly practices, incorrect treatment and painting, incomplete welds, rust, cracks, and other defects that could impair its operation or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the Supplementary Quality Assurance Provisions (SQAP) shown on the drawings. The drawings specify the characteristics requiring SQAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.5 and 6.6.1).

4.1.2 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction examination (see 4.3).
(b) Quality conformance examination (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 Preproduction examination. The power unit shall be examined as specified in 4.5. Presence of one or more defects shall be cause for rejection.

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4.4 Quality conformance examination. Each power unit shall be examined for the defects specified in 4.5. Presence of one or more defects shall be cause for rejection.

4.5 Examination procedure. The power units shall be examined for the following defects:

101. Any part or component not in accordance with the SQAP requirements as shown on the drawings.
102. Any dimension other than those identified on the SQAP not as specified.
103. Treatment and painting not as specified.
104. Components missing or not as specified.
105. Assembly not as shown on the drawings.
106. Workmanship not as specified.

4.6 Inspection of preparation for delivery.

4.6.1 Preproduction pack examination. The preproduction pack shall be examined for the defects specified in 4.6.2.3. Presence of one or more defects shall be cause for rejection.

4.6.2 Quality conformance examination of pack.

4.6.2.1 Unit of product. For the purpose of examination, a completed pack prepared for shipment shall be considered a unit of product.

4.6.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

107. Materials, containers, and procedures not as specified. Each incorrect material, container, and procedure shall constitute one defect.
108. Generator set not preserved as specified.
109. Canvas cover not installed as specified, when furnished.
110. Electrolyte not packed as specified.
111. Gasoline cans not coated with preservative and the excess preservative not drained as specified.
112. Fire extinguisher not removed from trailer bracket and packaged.
113. Ground cable not disconnected and secured as specified. Cable and terminals not matchmarked as specified.

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114. Repair parts and maintenance tools not preserved as specified.
115. Cargo trailer components not preserved as specified.
116. Packing not in accordance with the applicable figures for the level required.
117. Marking illegible, incomplete, incorrect, or missing.
118. Depreservation guides not prepared and attached to the power unit as specified.

5. PREPARATION FOR DELIVERY

5.1 Preproduction pack. The supplier shall furnish a preproduction pack for examination within the time frame specified to prove, prior to starting production packaging, that the applied preservation, packaging, packing, and marking comply with the preparation for delivery requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.7).

5.2 Preservation and packaging. Preservation and packaging shall be level A, B or C, as specified. Level A and B packaging of power units and components and intermediate packaging shall be in accordance with table I and figures 1, 2, and 3 and documents referenced thereon, and as specified herein.

5.2.1 Level A.

5.2.1.1 Generator set. The generator set shall be preserved in accordance with MIL-G-52684, level A. The generator sets shall be secured directly through the barrier material to the trailer using the required bolts in lieu of the wood base specified in table I and figure 1 of MIL-G-52684. The humidity indicator and inspection port shall be installed on an end or side of the cap in a location convenient for inspection of the humidity indicator.

5.2.1.2 Fire extinguisher. The fire extinguisher shall be removed from the bracket and packaged in a box conforming to PPP-B-636, type CF, variety SW, grade 125, style optional.

5.2.1.3 Cable. The ground cable to and terminal on the generator sets shall be matchmarked with cloth tags conforming to UU-T-81, size A, to facilitate reinstallation of the cable.

5.2.1.4 Repair parts. The preservation application criteria and applicable submethods of MIL-P-116 shall be used to preserve and package the repair parts.

5.2.1.5 Maintenance tools. Maintenance tools shall be preserved in accordance with PPP-P-40, level A.

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6.3 Preproduction model. Any changes or deviations of production power units from the approved preproduction model during production shall be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish power units conforming to this specification.

6.4 Government-furnished property. The contracting officer should arrange to furnish the property specified in 3.4.

6.5 Supplementary Quality Assurance Provisions (SQAP). The contracting officer should require the supplier to maintain records of all SQAP inspections. A suggested paragraph is as follows:

> "The supplier shall maintain complete records of all examinations and tests performed to verify the requirements of classified SQAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, corrected action taken when applicable."

6.6 Definitions.

6.6.1 Supplementary Quality Assurance Provisions (SQAP). A SQAP is a contractual requirement that supplements section 4 of the specification. The SQAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.7 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the supplier of his obligation to preserve, package, pack, and mark the power units in accordance with this specification.

6.8 Electrolyte. Detailed requirements for packaging of electrolyte for air shipment are contained in TM 38-250, "Packaging and Handling of Dangerous Materials for Transportation by Military Aircraft".

6.9 Depreservation guide. When applicable, the contracting officer should arrange to furnish the necessary copies of DA Form 2258, when requested by the supplier (see 5.5).

Custodian:

Army - ME

Preparing activity:

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Project No. 6115-A272

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TABLE I. PRESERVATION, PACKAGING AND PACKING MATERIAL

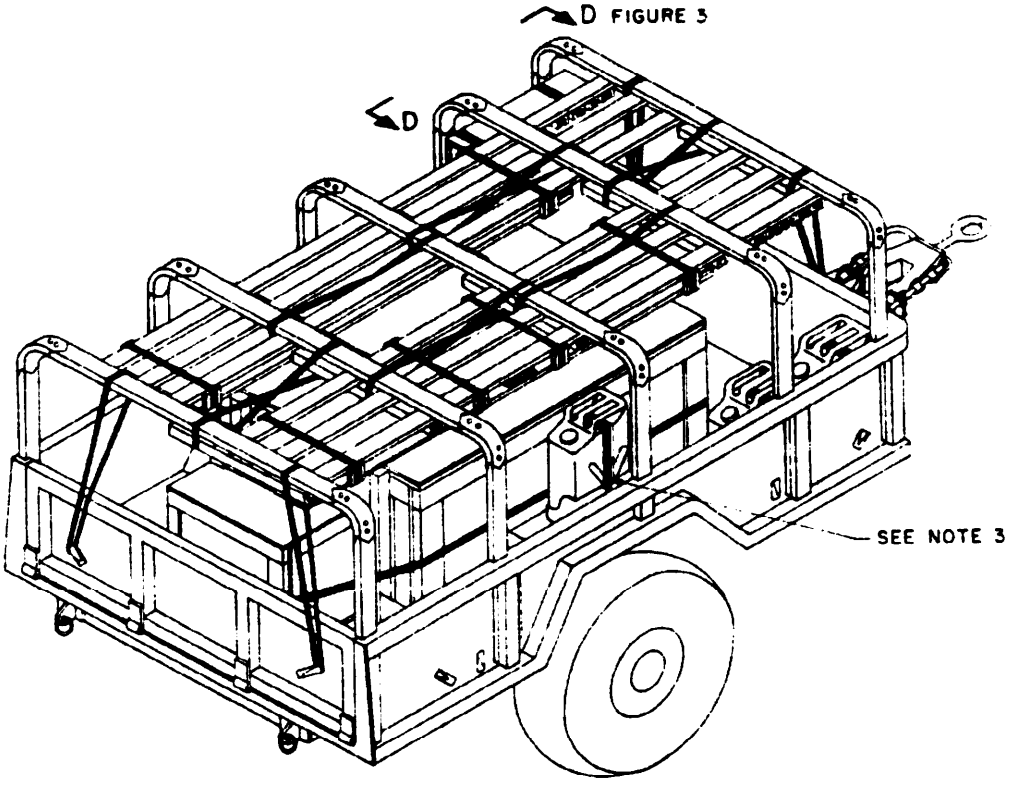
FIND NO.	IDENTIFICATION	DESCRIPTION	LEVEL		
			A	B	C
1	PPP-B-601, OVERSEAS TYPE, STYLE I (NOTES 1 AND 6)	BOX, PLYWOOD, $\frac{1}{4}$ THICK	1		
1	PPP-B-591, CLASS II, STYLE A (NOTES 1, 2, AND 3)	BOX, FIBERBOARD		1	
2	MIL-B-131, CLASS 1	BARRIER MATERIAL	AR		
3	PPP-F-320, TYPE SF, GRADE PS400	FIBERBOARD	AR		
4	MIL-G-12803, TYPE I, RBR BINDER, $1\frac{1}{2} \times 1\frac{1}{2} \times \frac{1}{8}$	GASKET, $\frac{11}{16}$ DIA HOLE IN CENTER	8		
5	WEDGE	LUMBER, SOFTWOOD	2	2	
6	MS27183-11, .312 ID X .734 OD X .065 THK	WASHER, PLAIN, CS, CD PL	6	6	
7	MS16992-507, $\frac{1}{4}$-10 X 3 LG	BOLT, LAG, HEX HD, GIMLET PT, STL, GALV	6	6	
8	PPP-B-601, OVERSEAS TYPE, STYLE I, TYPE I LOAD	BOX, PLYWOOD	1		
8	PPP-B-601, DOMESTIC TYPE, STYLE I, TYPE I LOAD	BOX, PLYWOOD		1	
9	BLOCKING	LUMBER, SOFTWOOD, 1 X 2 NOMINAL	AR	AR	AR
10	L-P-378, TYPE I, CLASS 1, GRADE A, 4-8 MIL	PLASTIC FILM, POLYETHYLENE	AR	AR	AR
11	BLOCKING	LUMBER, SOFTWOOD, 2 X 2 NOMINAL	3	3	
12	PPP-T-60, TYPE IX, CLASS 1	TAPE, PRESSURE SENSITIVE	AR	AR	AR
13	PPP-T-97, TYPE IX	TAPE, PRESSURE SENSITIVE, $\frac{3}{4}$ WIDE	AR	AR	
14	QQ-S-781, TYPE I OR IV, CLASS B	STEEL STRAPPING, FLAT, .035 X $\frac{3}{4}$ WIDE	AR		
14	QQ-S-781, TYPE I OR IV, CLASS A	STEEL STRAPPING, FLAT, .035 X $\frac{3}{4}$ WIDE		AR	AR
15	FF-N-105, TYPE II, STYLE 4, TABLE Vb	NAIL, 10d	AR	AR	
16	NN-P-530 (NOTE 5)	PLYWOOD, $\frac{1}{4}$ THICK		AR	AR
16	NN-P-530 (NOTE 4)	PLYWOOD, $\frac{1}{4}$ THICK	AR		

NOTES:

1. NO CONTENT WEIGHT, WITHOUT BOTTOM.
2. TOP TO BE INVERTED.
3. DIMENSIONAL LIMITATIONS OF SPECIFICATION WAIVED.
4. GROUP A, CONFORMING TO CS-35, TYPE I, GRADE 3-4 OR GROUP B, CONFORMING TO PS-1 STANDARD SHEATHING WITH EXTERIOR GLUE.
5. GROUP A, CONFORMING TO CS-35, TYPE I, GRADE 3-4 OR GROUP B, CONFORMING TO PS-1 STANDARD INTERIOR SHEATHING.
6. TOP TO BE ONE PIECE, 1/2 THICKNESS, AND SHALL COMPLETELY COVER ALL EDGES.

DX-1274B

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NOTES

1. SEE TABLE I FOR LIST OF MATERIAL.
2. HAMMER, FUEL DRUM ADAPTERS, EXTENSION, FIRE EXTINGUISHER, GROUND RODS, OIL DRAIN, GROUND WIRES, AND ATTACHING HARDWARE SHALL BE SECURED TO THE GENERATOR SET, USING TAPE, FIND NO 13.
3. 5 GALLON GAS CANS SHALL BE SECURED IN THEIR NORMAL POSITIONS USING STRAPPING, FIND NO 14, AROUND OUTSIDE OF GAS CAN BRACKETS
4. TARPAULIN SHALL BE WRAPPED IN FIND NO.10 AND SECURED WITHIN THE CONFINES OF FIND NO. 8

FIGURE 1.
PACKING ARRANGEMENT AND DETAIL
DX-1277A

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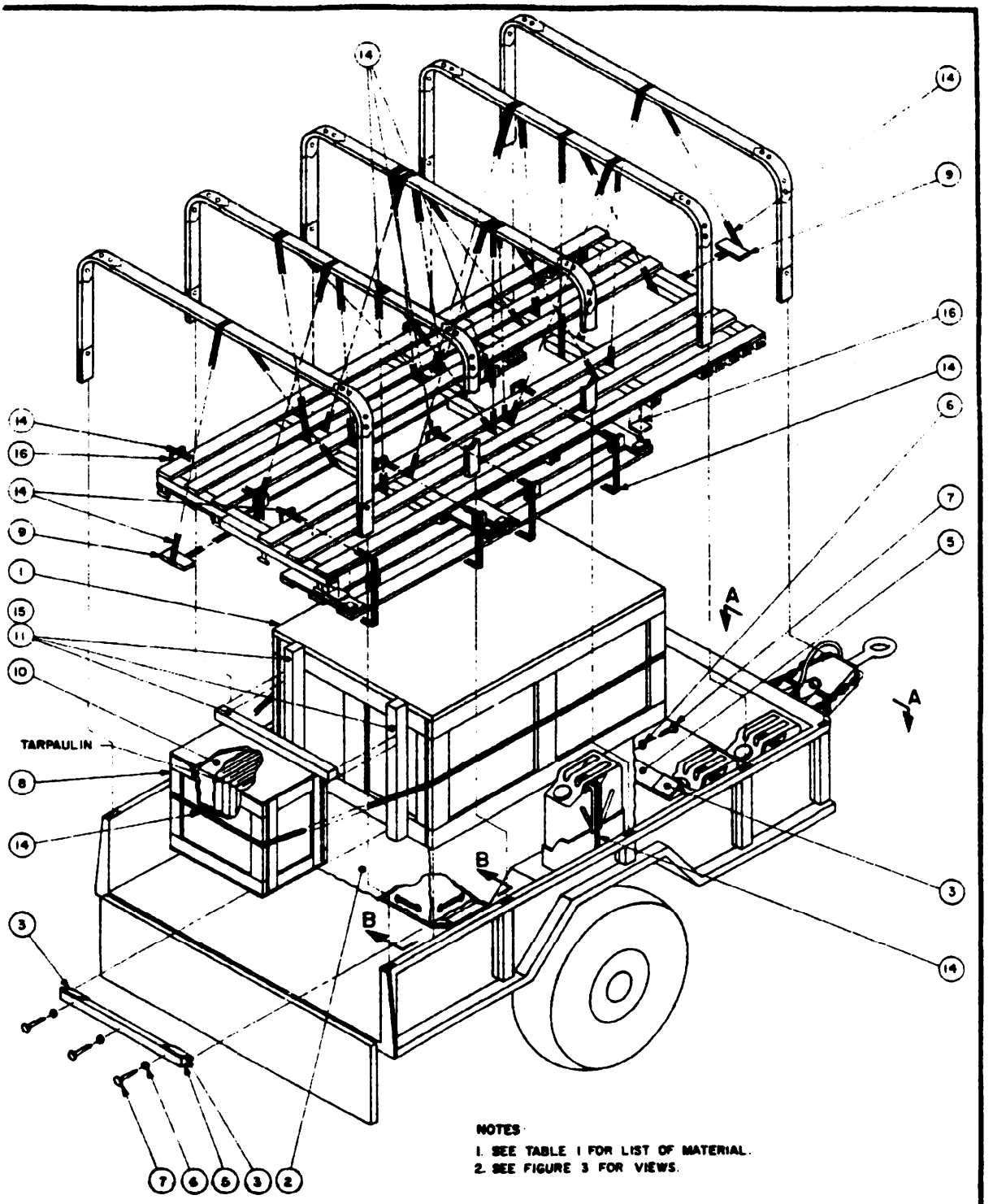


FIGURE 2. PACKAGING AND PACKING ASSEMBLY DX-1275A

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5.2.3 Level C.

5.2.3.1 Generator set. The generator set shall be preserved in accordance with MIL-G-52684, level C. Cable and oil drain lines shall not be disconnected. When furnished, the canvas cover shall be installed.

5.2.3.2 Fire extinguishers, repair parts, maintenance tools and technical publications. The fire extinguishers, repair parts, maintenance tools and technical publications shall be preserved and packaged as specified for level A.

5.2.3.3 Other components. Components not covered by specific requirements shall be preserved and packaged to afford protection against deterioration and damage during shipment and handling from the power unit supplier to the initial shipping destination.

5.2.3.4 Electrolyte. For air shipment, the electrolyte shall be overpacked and secured as specified for level A.

5.3 Packing. Packing shall be in accordance with table I and figures 1, 2, and 3 for level A, B, or C as specified.

5.4 Marking. Marking shall be in accordance with MIL-STD-129.

5.5 Depreservation guides. Two copies of depreservation guides shall be prepared and placed in waterproof envelopes marked "Depreservation Guide". One copy shall be secured on one of the generator sets and a second copy shall be secured in a conspicuous location on the power unit. Unless otherwise specified, DA Form 2258 "Depreservation Guide for Vehicles and [illegible] shall be used (see 6.9).

TES

Intended use. The power unit is intended for use by the armed services for supplying electrical power to military operations.

Ordering data. Procurement documents should specify the following:

(a) Title, number, and the date of this specification.
(b) Time frame required for submission of preproduction model (see 3.2).
(c) Time frame required for submission of the preproduction pack (see 5.1).
(d) Level of preservation and packaging and level of packing required, (see 5.2 and 5.3). Level B preservation and packaging is intended to provide economical but limited protection and should be specified only when it has been determined that the power units will be held in covered storage for an indefinite period or when the anticipated shipment and storage time will exceed 30 days.
(e) When other than DA Form 2258 is to be used (see 5.5).

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5.2.1.6 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1.

5.2.1.7 Electrolyte. When furnished, the electrolyte shall be overpacked in a wood box and secured to the power unit as shown on the figures. When air shipment is required, the packaged electrolyte units within the overpack container shall be cushioned on all sides with noncombustible absorbent material in sufficient quantity to completely absorb liquid contents in event of breakage. The box shall be secured to the power unit in any available space that will not increase the overall cube of the power unit (see 6.8).

5.2.1.8 Cargo trailer. The cargo trailer components shall be preserved in accordance with MIL-STD-281, level A, as specified for mobile vehicles. Brake components shall be coated with primer as specified therein.

5.2.1.9 Gasoline cans. Each gasoline can shall be flushed or fogged with preservative lubricating oil conforming to MIL-L-21260, grade 2. Excess preservative shall be drained. The cans shall be secured in the mounting bracket.

5.2.2 Level B.

5.2.2.1 Generator set. The generator set shall be preserved in accordance with MIL-G-52684, level B.

5.2.2.2 Fire extinguisher. The fire extinguisher shall be removed from the bracket and packaged as specified for level A.

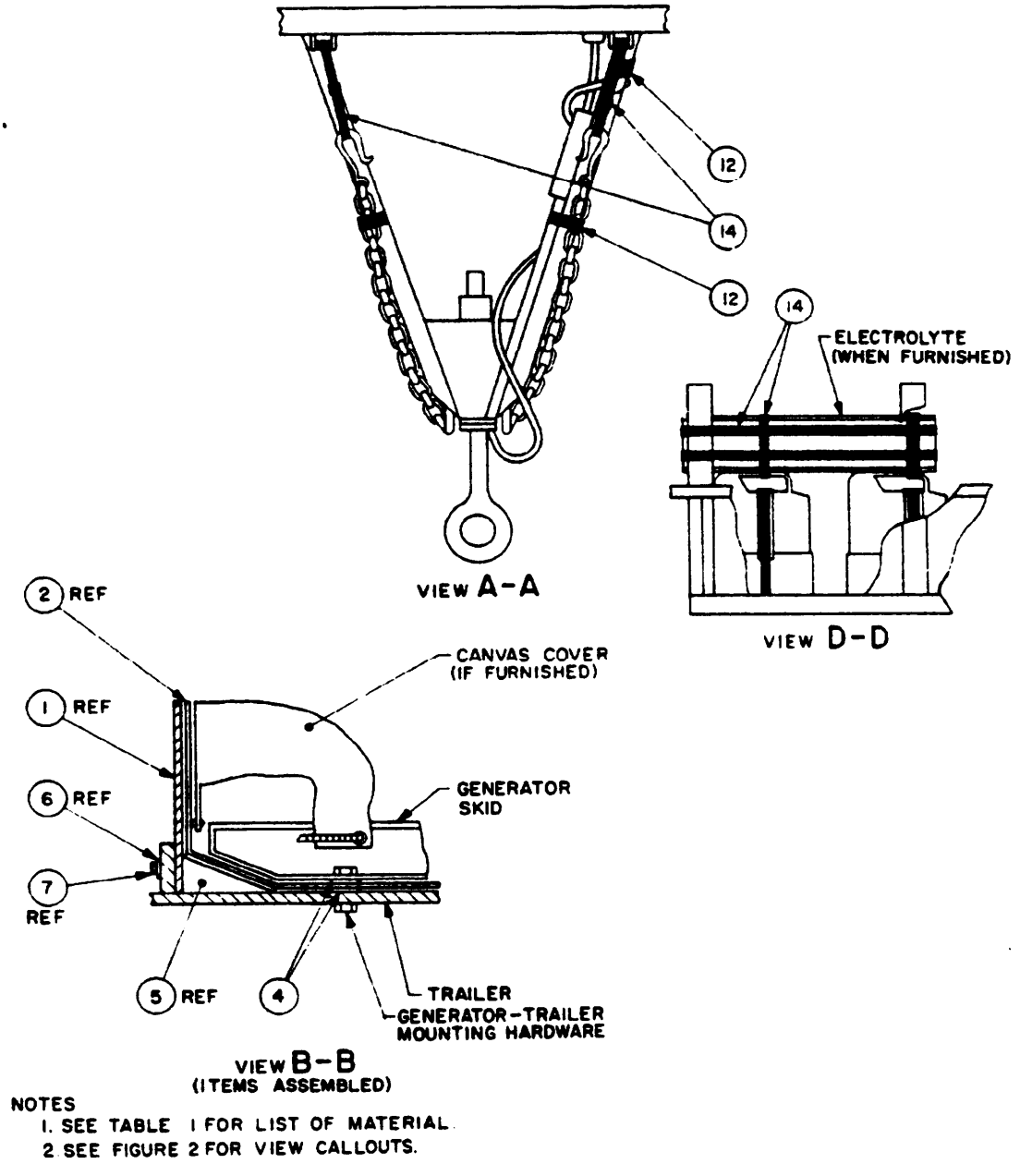
5.2.2.3 Cable. Matchmarking information shall be as specified for level A.

5.2.2.4 Repair parts, maintenance tools and technical publications. The repair parts, maintenance tools and technical publications shall be preserved as specified for level A.

5.2.2.5 Electrolyte. When furnished, electrolyte shall be overpacked and secured as specified for level A.

5.2.2.6 Cargo trailer. The cargo trailer components shall be preserved in accordance with MIL-STD-281, level B, as specified for mobile vehicles.

5.2.2.7 Gasoline Cans. Gasoline cans shall be preserved as specified for level A.



NOTES

1. SEE TABLE I FOR LIST OF MATERIAL.
2. SEE FIGURE 2 FOR VIEW CALLOUTS.

FIGURE 3. PACKAGING AND PACKING ASSEMBLY

DX-1276A

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION: MIL-G-52627A(ME) - GENERATOR SET, GASOLINE ENGINE, TRAILER MOUNTED, 5 KW, 60 HERTZ, PU-409A/M

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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